

Blotting techniques allow for the detection and characterization of a specific biomolecule of interest within a sample of multiple biomolecules. The first blotting technique to be widely used was published in 1975 by the biologist Edwin M. Southern. This technique detects specific DNA sequences and bears the name of its creator: the Southern blot. Just two years later in 1977, James Alwine, David Kemp, and George Stark published a similar method to detect specific RNA sequences. In homage to Southern's technique, this method was named the northern blot. The western blot, which uses antibodies to detect specific proteins, followed in 1981.

Since then, other blots that are less well known have been developed as slight modifications of these first three methods. These include the eastern blot, which detects post-translational modifications; the southwestern blot, which detects protein–DNA interactions; and the northwestern blot, which detects protein–RNA interactions.

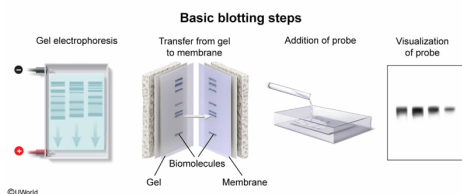
Each of the blotting techniques mentioned in the passage most likely requires which of the following steps?

- ☐ A. Purification of target molecules by chromatography [14%]
- ✓ ☒ B. Separation of molecules by gel electrophoresis [66%]
- ☐ C. Hybridization of target molecules to a nucleic acid probe [7%]
- ☐ D. Digestion of large molecules into smaller fragments [12%]

Correct

66% answered correctly

Explanation:



Blotting techniques include the [Southern](#), [northern](#), and [western](#) blots, along with other derivatives, and involve the **separation of similar biomolecules** (eg, multiple DNA molecules of different size) followed by the **detection of a specific biomolecule** of interest. These techniques are powerful because they can **detect specific molecules within a mixture without extensive prior purification**. Each blotting technique involves the following general steps:

1. A mixture of molecules is subjected to [gel electrophoresis](#) to be separated by size. Typically, agarose gels are used for nucleic acids and polyacrylamide gels are used for proteins, though agarose gels may be used to separate large proteins and polyacrylamide gels may be used for small nucleic acids.
2. The separated biomolecules are transferred from the gel to a membrane such as nitrocellulose or polyvinylidene difluoride.
3. The molecules are exposed to probes that specifically bind to the target molecule. For nucleic acids, the probe is typically a complementary nucleic acid, whereas for proteins the probe is typically an antibody. For western blots, the membrane must first be [blocked](#) with milk or bovine serum albumin (BSA) so that antibodies do not bind nonspecifically to the membrane itself.
4. The probe is visualized, often by fluorescence, autoradiography, or chemiluminescence.

The question asks for the step that is most likely required for each blotting technique mentioned in the passage. Of the given answer choices, separation of molecules by **gel electrophoresis** is the only choice that is **common to all the blotting techniques mentioned in the passage**.

(Choice A) Chromatography separates molecules on the basis of one or more properties such as size, charge, or affinity for a ligand, and allows for the purification of a molecule of interest. Blotting techniques typically do *not* require purification by chromatography.

(Choice C) Some blotting techniques require hybridization of molecules of interest to a nucleic acid probe, but others (eg, western blots) use probes that are *not* nucleic acids.

(Choice D) DNA is often digested into smaller fragments prior to gel electrophoresis, but RNA and proteins are typically left intact.

Educational objective:

Blotting techniques involve the separation of biomolecules by electrophoresis and the detection of a specific molecule using a specific probe.

Time spent:0 sec

Subject:
Biochemistry

QID:404137

System:
Amino Acids and Proteins

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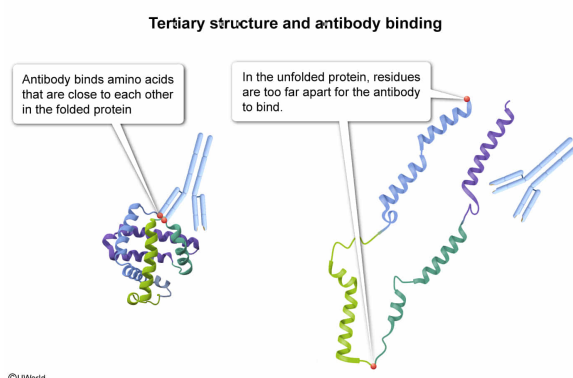
In which of the following scenarios should sodium dodecyl sulfate NOT be used during the generation of a western blot?

- ✔ ☒ A. When the epitope of interest involves amino acids that are far apart in the primary structure [30%]
- ☐ B. When the epitope of interest involves amino acids that are adjacent to each other in the primary structure [28%]
- ☐ C. When the protein of interest has an isoelectric point above physiological pH [19%]
- ☐ D. When the protein of interest has an isoelectric point below physiological pH [20%]

Correct

31% answered correctly

Explanation:



Western blots use antibodies as probes to detect specific proteins. A given antibody typically binds to a specific region within a protein, known as an **epitope**. This binding interaction results from **noncovalent forces** between amino acid residues in the antibody

and amino acid residues in the protein of interest.

Many epitopes consist of several amino acids that are adjacent to each other in the **primary structure** of the protein of interest. An appropriate antibody can bind an epitope of this sort regardless of whether the protein is properly folded. In contrast, **some epitopes consist of amino acids that are far apart from each other** in the primary structure of the protein but are **brought close together in its tertiary structure when the protein folds**. For epitopes of this kind, **unfolding the protein prohibits the antibody from binding**.

Sodium dodecyl sulfate (SDS) is a **denaturing agent** commonly used in **gel electrophoresis** of proteins (ie, SDS-PAGE). As a denaturing agent, **SDS** causes proteins to unfold and therefore **should not be used** in western blots for which the **epitope consists of amino acids that are far apart** in the primary structure.

(Choice B) Epitopes that consist of amino acid residues that are adjacent to each other in the primary structure of a protein can be **bound by an antibody** regardless of whether the protein is properly folded, so use of SDS during the electrophoresis portion of the western blot does *not* affect the outcome.

(Choices C and D) The **isoelectric point** (pI) of a protein determines its charge at a given pH: if the pI is below the pH, the protein is negatively charged, and if the pI is above the pH, the protein is positively charged. However, SDS uniformly coats proteins of any pI with negative charges and may be used in a western blot regardless of the pI of the protein of interest.

Educational objective:

Antibodies bind specific portions of proteins called epitopes. If the amino acids in an epitope are far apart in the protein's primary structure, then denaturing the protein prevents antibody binding.

Time spent:0 sec

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Biochemistry

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If an eastern blot is used to detect phosphorylation, it will most likely detect modifications to which of the following amino acid residues?

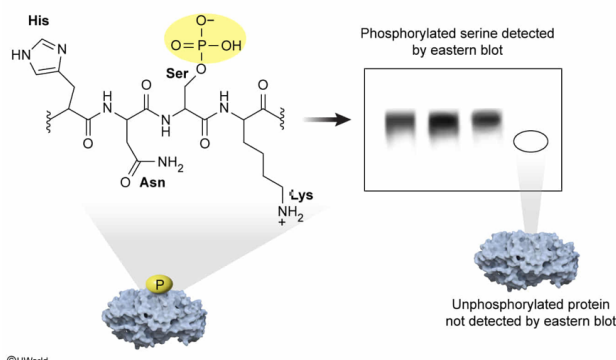
- ☐ A. His [10%]
- ☐ B. Asn [5%]
- ✓ ☒ C. Ser [74%]
- ☐ D. Lys [9%]

Correct

74% answered correctly

Explanation:

Serine, threonine, and tyrosine are the most common phosphorylation sites



Post-translational modifications are additions of non-amino acid chemical groups

to proteins. Common modifications include phosphorylation, glycosylation, and ubiquitination. Generally, a specific type of modification will occur on the side chain of only one amino acid species or a few side chains with similar **functional groups**.

According to the passage, eastern blots are used to detect post-translational modifications. The question asks specifically about the detection of protein phosphorylation. Typically, phosphorylation occurs when the **hydroxyl group** of an amino acid side chain acts as a **nucleophile** to attack a phosphate group on ATP (this reaction is facilitated by a **kinase enzyme**). As such, the **most common** sites of phosphorylation are the residues that contain hydroxyl groups: **serine** (Ser), **threonine** (Thr), and **tyrosine** (Tyr). Other sites such as **glutamate** (Glu) are occasionally phosphorylated, but this is rare. Therefore, of the given choices, an eastern blot is **most likely to detect phosphorylation of a serine (Ser) residue**.

(Choice A) **Histidine** (His) is occasionally phosphorylated in certain biochemical pathways, but this is rare, so His is *less likely than Ser* to be detected as a phosphorylation site.

(Choice B) **Asparagine** (Asn) is a common site for **glycosylation**, but not phosphorylation.

(Choice D) **Lysine** (Lys) is the most common site of **ubiquitination**, but it is rarely phosphorylated.

Educational objective:

Phosphorylation is a post-translational modification that most commonly occurs on the hydroxyl groups of serine, threonine, and tyrosine residues.

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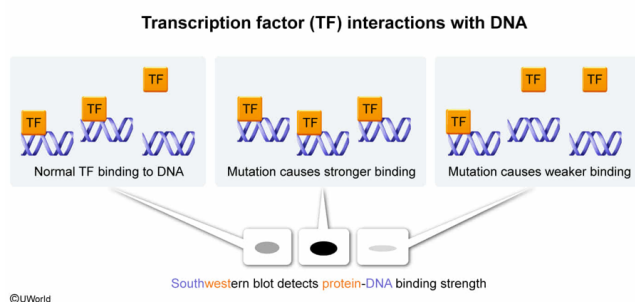
Which of the techniques described in the passage would be best for probing the effect of a mutation on the affinity of a transcription factor for its ligand?

- ☐ A. Southern blot [7%]
- ☐ B. Northern blot [7%]
- ✓ ☒ C. Southwestern blot [63%]
- ☐ D. Northwestern blot [20%]

Correct

63% answered correctly

Explanation:



Transcription factors (TFs) are proteins that **bind to specific DNA sequences** to regulate **transcription** (ie, RNA synthesis) of the corresponding gene. Binding of a TF to DNA may either **upregulate or downregulate transcription**.

The question asks for the technique from the passage that would best probe the effect of a mutation on the binding of a TF to its **ligand**. For a TF, the ligand is the DNA sequence of interest. Note that the mutation may alter either the amino acid sequence of the TF or the nucleotide sequence of the ligand. Either change could affect binding affinity.

The passage states that southwestern blots detect protein–DNA interactions. Therefore, of the available choices, the **southwestern blot would be best for determining the effect of a mutation on binding affinity** between a TF (a protein) and its ligand (DNA).

(Choice A) **Southern blots** detect specific DNA sequences, but they do not detect interactions between those sequences and proteins.

(Choice B) **Northern blots** detect specific RNA sequences. This technique can be used to detect changes in transcription levels in response to a mutation. However, because TF binding can either upregulate or downregulate transcription, northern blots do not tell whether the mutation caused increased or decreased TF binding to DNA.

(Choice D) According to the passage, northwestern blots detect protein–RNA interactions. TFs bind *DNA*, not RNA.

Educational objective:

Transcription factors are proteins that bind DNA and either increase or decrease transcription levels of one or more genes.

Time spent:0 sec

Subject:
Biochemistry

QID:404140

System:
Amino Acids and Proteins